

src/simulator_interface.py

48 lines

```
1 """
2 The contract between the controller and *any* well simulator.
3
4 The whole point of this module is to make one claim provable rather than
5 merely asserted:
6
7 The controller never touches simulator internals. It only ever calls
8
9 Q, WHP, FLP, BHP = simulator.step(choke_position)
10
11 so the official Honeywell simulator can be dropped in place of the
12 physics stand-in in `well_simulator.py` with ZERO controller changes.
13
14 `scripts/04_run_scenarios.py` type-checks the simulator against this Protocol
15 before every run, and `tests/` exercises the controller against a deliberately
16 different dummy plant that satisfies the same Protocol.
17 """
18
19 from __future__ import annotations
20
21 from typing import Protocol, Tuple, runtime_checkable
22
23
24 @runtime_checkable
25 class WellSimulator(Protocol):
26     """Minimal interface required of a well simulator."""
27
28     def step(self, choke_position: float) -> Tuple[float, float, float, float]:
29         """
30         Advance the plant by one control interval (Ts = 1 hour).
31
32         Parameters
33         -----
34         choke_position : float
35             Commanded production choke opening in percent, 0-100.
36
37         Returns
38         -----
39         (Q, WHP, FLP, BHP) : tuple of float
40             Oil flow rate [bbl/hr], wellhead pressure [psi],
41             flowline pressure [psi], bottom hole pressure [psi].
42         """
43         ...
44
45     def reset(self) -> Tuple[float, float, float, float]:
46         """Return the plant to its initial (shut-in) condition."""
47         ...
48
```